

LIST OF CHANGES AND RESPONSES TO THE REFEREE

We extend sincere gratitude to the referee for their comments and careful eye. We have edited our paper to incorporate these comments. We list the referee's comments below. Our responses are *italicized*.

Comments and Responses.

- (1) p.1, 12th line of the introduction: using *a* cohomological argument:
Fixed
- (2) p.5, l.19: The comma should be a period. *Fixed*
- (3) p.6, l.1-18: Is it clear that this dimension is not 0, or is it part of Theorem 1.3? *Clarified that the dimension of this space is nonzero (as long as the graph G_0 has nontrivial fundamental group, since each of the morphisms is a homotopy equivalence) and is bounded above by the $\liminf$.*
- (4) p.6, l.-7: When you write “putting a metric on G_s ”, you probably want to require some compatibility: should this be a length measure on the sequence G_s (and relatedly to my previous question, do you know in advance that this exists)? *Clarified this by changing “putting a metric on G_s ” to “assigning a length measure to the folding sequence $(G_s)_{s \geq 0}$ ” Is the “natural linear isomorphism” in Theorem 1.3 the map sending the length measure on (G_s) to the length measure on T via this construction? Yes, added clarification of this point to the sentence before Theorem 1.3.*
- (5) I have the same questions for width measures: is it clear that they always exist, and is there a way of describing the “natural linear isomorphism” in Theorem 1.4? *We added a sketch of the proof of [NPR14, Theorem 4.4], constructing a current on the legal lamination from the width measures on a reduced unfolding sequence.*
- (6) p.6-7: For the benefit of the reader and clarity of exposition, it might be worth recalling what you mean by a *topological F_n -tree* and by a *length measure* on an F_n -tree. *We added the definitions of topological F_n -tree (locally path connected, uniquely path connected space admitting an F_n action) and the definition of a length measure on an F_n -tree (a compatible, equivariant collection of finite Borel measures on compact intervals of T).*
- (7) p.7, l.7: This is Proposition 5.4 in [26] *Changed to say “Proposition 5.4” in [26]*
- (8) p.8, l.15: Missing space between tree and T *Fixed*

- (9) p.10, l.10: The intersection form should be defined on $\overline{cv_n}$ rather than $\overline{CV_n}$, unless you should a continuous lift $\overline{CV_n} \rightarrow \overline{cv_n}$. Same two lines below. *Yes, you are absolutely correct. We have made this change, and also changed the language in the following paragraph to reflect this change. We also added clarification that the set T^* is scale-invariant, which by Lemma 1.7(ii) implies that arationality is a property of the scalar class of a tree. This then allows us to only consider arational trees in $\overline{CV_n}$.*
- (10) p.10, l.14: Why does this unique minimal invariant set exist? Do you need to require that it contains a certain current? *The minimal closed invariant subset of the current space exists, and this is discussed in Section 5.8 of Reiner Martin's thesis [Mar95]. We added the reference as a clarification in the manuscript.*
- (11) p.11, l.-13: You might want to add the word *reduced* in front of *folding sequence*. *Added that clarification to that sentence, as well as to the sentence following.*
- (12) p.11, l.-3: I was slightly confused by the logic in your formulation "In particular, by Proposition 1.6,..." Are you implying that convergence of the volume to 0 is a consequence of the above combined with Proposition 1.6, and if so, why? *We agree this was confusing. As that sentence is not needed for the proof, we deleted the sentence to avoid confusion.*
- (13) p.12, l.-21: Typo on trees. *Fixed*
- (14) p.12, l.-7: Remove an occurrence of the word of. *Fixed*
- (15) p.13, l.12: Missing space before the parenthesis. *Fixed*
- (16) p.13, l.20: I do not understand why you are specifically pointing to point (ii) of Lemma 1.7, here. *You are correct - we have already used that T and T' lie in the same simplex in ∂CV_n earlier in this proof, so we do not need to call upon Lemma 1.7(ii) here. We have therefore removed this reference.*
- (17) p.14, l.6: I got confused by the sentence "There is a decreasing sequence t_i such that every v_{t_i} is a vertex": do you mean that you could have chosen v_{t_i} to be a vertex? (The sentence could be interpreted as: all preimages of v_t in G_{t_i} are vertices, since you just chose a preimage arbitrarily in the previous paragraph.) *We changed "There is a decreasing sequence $t_i \rightarrow -\infty$ so that every v_{t_i} is a vertex" to "Then, we can pass to a subsequence t_i with $t_i \rightarrow -\infty$ such that each v_{t_i} is a vertex of G_{t_i} " to clarify this sentence.*
- (18) p.14, l.-13: Do you know when $\mathbb{PC}(\Lambda)$ is a finite-dimensional simplex? As a stupid example, for a constant unfolding sequence, $\Lambda = \partial^2 F_n$ and $\mathbb{PC}(\Lambda) = \mathbb{PCurr}(F_n)$. *This is true for arational limiting tree, as by Theorem 1.14 we can find a reduced unfolding sequence limiting to it. This is because the inverse limit of simplices of bounded dimension*

has bounded dimension. The constant unfolding sequence that referee gave is not reduced.

- (19) p.20, l.8: It is a bit unfortunate to have a sequence of $n \times n$ matrices called A_n . Same at the beginning of Section 3.2 on page 27. *Happy to make this change! We changed the index variable of the matrices from n to j both here and on page 27.*
- (20) p.20, l.-3: Theorem 1.3 has an assumption that the folding sequence is reduced, which you do not have here. Is it not needed? Same for Theorem 3.5, which relies on Theorem 1.4. *Yes and yes. We added reduced in the statements for Theorems 1.4, 1.5, 2.4, and 3.5.*
- (21) p.21, Theorem 2.5: Do you want to state (and briefly explain in the proof) that the sequence you use is reduced? (See also my previous comment.) Otherwise, formally speaking, am I wrong to think that the theorem trivially holds true by taking a constant folding sequence on a simplicial tree with at least $2n - 6$ edges? Same for Theorem 3.6. *Yes! We made this change in the theorem statement for 2.5 and added justification at the end of the proof of Theorem 2.5 for why the folding sequence is reduced. We also added the word reduced to the statement of Theorem 3.6 and adjusted the proof accordingly.*
- (22) p.28, l.12: Should this $\delta_{r,s}$ be $p_{r,s}$? *Yes, fixed*
- (23) p.31, l.-4: Your choice of notation is a bit unfortunate: you now use $\phi_{s,0}$ for the morphism, which used to be $f_{s,0}$ in Section 1, but then you have a conflict of notation with the edge f_s on p.31, l.-1. *We have replaced the notation $f_s : G_s \rightarrow G_{s+1}$ to $\psi_s : G_s \rightarrow G_{s+1}$ and we introduce the notation $\phi_{r,s}$ right after.*
- (24) p.33, l.1: One of these should be $\ell_r(e')$ *Fixed*
- (25) p.33, l.5: You write “By the previous paragraph”, but I am not sure which paragraph you are referring to: is it not a consequence of the previous sentence? *You are correct, we deleted this phrase - it does follow from the previous sentence.*
- (26) p.35, l.10: I got confused by the phrasing of your argument, here. Are you implying that any ergodic measure is *supported* on some B_j^i ? In what sense? In the sense that every other edge has measure 0, or in a weaker sense? *Yes, and we meant that an edge e supports an ergodic measure ℓ^i or μ^i if $e \in H^i$ for the corresponding transverse decomposition described in Theorems 1.16 and 1.17. We added a definition of what it means for an edge to support a measure thereafter.*
- (27) p.35, Theorem 5.1: I recommend writing *length measures* instead of just *measures* to avoid any possible confusion. *Fixed*
- (28) p.36: In the caption of Figure 5, there seems to be a missing space between turns and the parenthesis. *Fixed*
- (29) p.37-38: I found the proof of Lemma 5.3 hard to read, perhaps because it is quite descriptive and requires the reader to complete the computations on their own. For example, what do you mean when

you say “the Alice-Bob game still holds”? What precise conditions allow you to insert a folding map inside your original folding sequence, ensuring that the conclusion from the theorem in Section 2 still holds? Is one implicitly required to compute again the new transition matrix, or is there a general lemma saying that inserting a matrix with bounded coefficients in the product does not affect the desired conclusion? Is it important to know that the times at which you insert new folds are disjoint for distinct free factors H_1, H_2 ? I generally felt that having a few extra details throughout the proof would be helpful to follow the argument. *We added a sentence to the beginning of Section 2.2 clarifying that the goal of the Alice-Bob game is to construct a folding sequence whose transition matrices will satisfy the conditions in Theorem 2.4. Also added the reference to the Alice’s winning conditions as List 1 and List 2. We also clarified key steps of the proof to increase readability and add in additional details. In response to the questions you asked: Yes, implicitly we are required to recompute the new transition matrix, which we clarified in the proof, however a result of this is that yes, adding in a matrix with bounded coefficients in the product does not affect the desired conclusion. As for the different free factors: you could worry that there is a point in time where multiple folds are inserted at the same time in the folding sequence for different free factors that could interact with one another. However, this will never happen since we only consider one free factor at a time, and once that free factor is taken care of, we continue to fold in order to then take care of the next free factor. Once the illegal turns become legal for a single free factor, it stays legal forevermore, even when new folds are introduced. We added clarification of this point at the end of the proof.*

- (30) A general question: I was wondering if you knew what happens to your main theorems if you do not require that the limiting trees are arational, but something weaker, e.g. that they have dense F_n -orbits? In the surface setting, filling laminations seem to realize the maximum possible number of transverse measures, should one expect something similar for free groups (not quite because of simplicial trees with roughly $3n$ edges)? *That’s a good guess; there are lemmas (Such as Lemma 1.15) for arational trees which could be improved for trees with dense orbits; NPR requires the sequences to be reduced; this is not automatic for trees with dense orbits. We think the mixing condition on the limiting tree is equivalent to the reduced condition on the folding sequence in which case the proof of the upper bound would go through for these trees.*

REFERENCES

- [Mar95] Reiner Martin. *Non-uniquely ergodic foliations of thin-type, measured currents and automorphisms of free groups*. ProQuest LLC, Ann Arbor, MI, 1995. Thesis (Ph.D.)—University of California, Los Angeles.
- [NPR14] Hossein Namazi, Alexandra Pettet, and Patrick Reynolds. Ergodic decompositions for folding and unfolding paths in outer space. *arXiv preprint arXiv:1410.8870*, 2014.